SpaceWire

Space Software

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ΕΛΛΗΝΙΚΗ ΔΗΜΟΚΡΑΤΙΑ
Εθνικό και Καποδιστριακό
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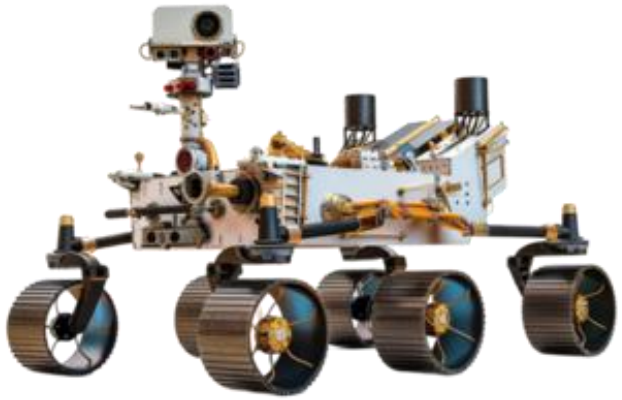
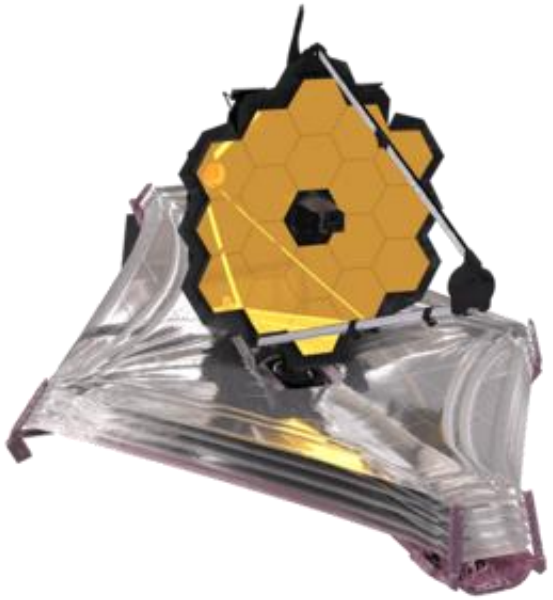
Data Transfer in Space

- Modern Spacecraft generate very large amounts of data in very high data rates
  - e.g. High Resolution cameras, Radar, etc.
- Data that needs to be moved efficiently between sources and mass-memory devices
- Operating in space introduces special challenges
  - Need Small Size / Weight
  - Need Low Power Consumption
  - Need Radiation/Fault Tolerance
- Interoperability / Modularity
  - Components come from many maufacturers





About SpaceWire

- Published by European Cooperation for Space Standardization
  - ECSS-E-ST-50-12 Rev 1 (2003)
- Has been designed into hundreds of missions
  - NASA, ESA, JAXA Roscosmos
  - Both Scientific and Commercial vehicles
- Developed specifically for spacecraft
  - Provides a standard communication protocol
  - Helps address many of the special challenges that come with operating in space

Technical Characteristics of SpaceWire

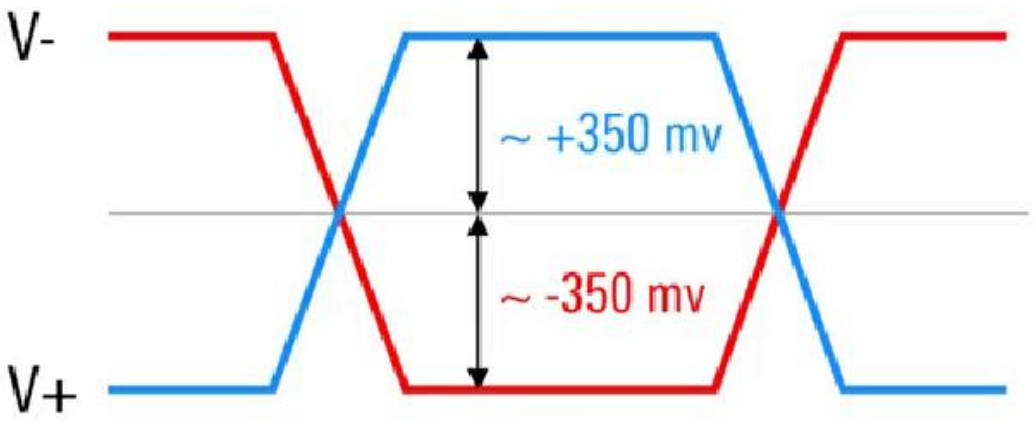
- Full Duplex
- Bidirectional Communication (~2-200Mbps)

- 2 Signals - Data & Strobe
- Cables - 4 Pairs of conductors + Shield
- Link initialization and error detection/recovery
- Data is sent in packets
    - P2P links or routed through a network
- Control & Time / Sync info can also be sent

LVDS

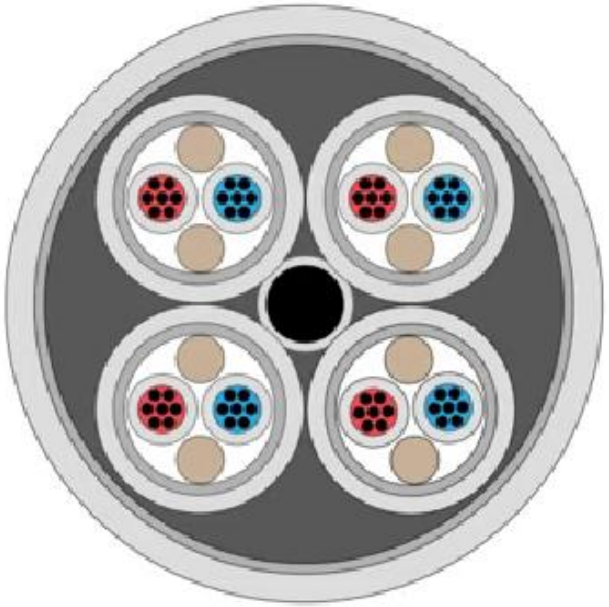
Low
Voltage
Differential
Signaling

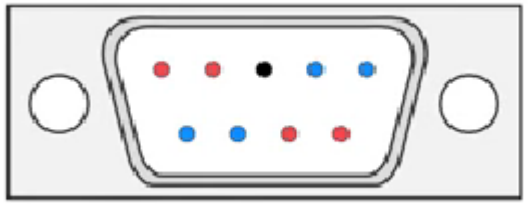
- Used for applications that need high speed but low power & noise
- Differential Signaling improves noise immunity and reduces EMI.
- Low Voltage refers to the small voltage swing with respect to the GND (~350 mV)



Cabling

- Cables contain 4 twisted pairs.
  - Each pair carries a differential signal
  - Each pair has its own shield
- Impedance: 100 Ohms
- Maximum Cable Length: 10 Meters
  - Can be extended with routers
- 9 Pin Type D Connector
  + 8 pins for diff pairs, 1 for shield
- Shielding each pair helps reduce crosstalk



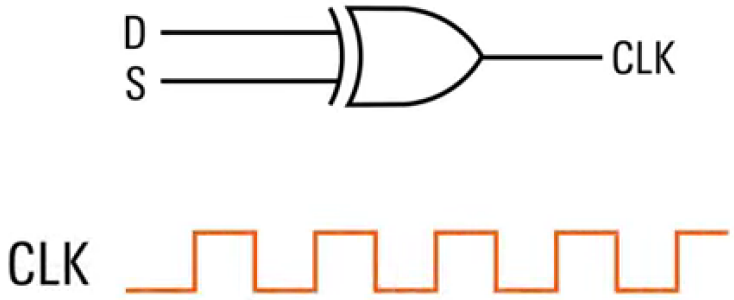


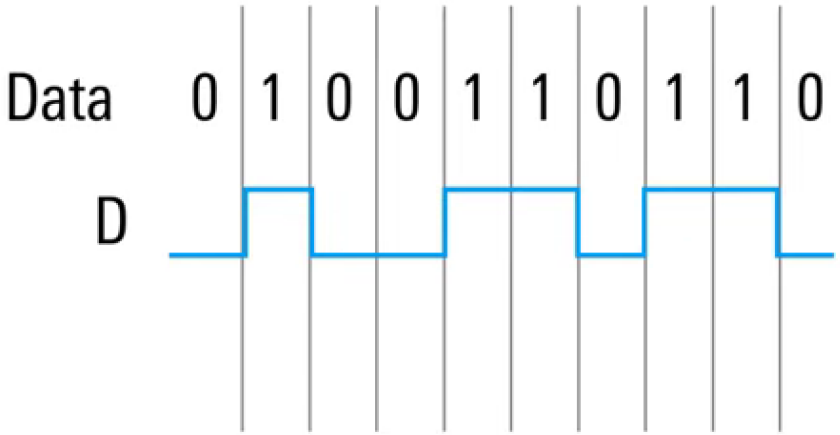
Cabling

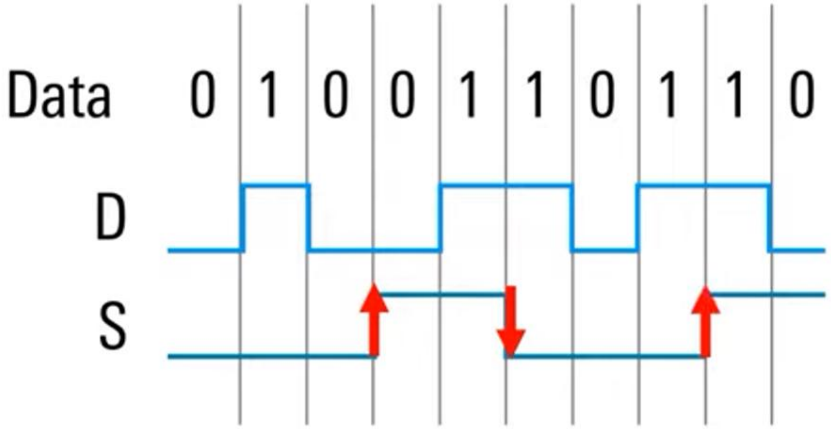
- 4 Differential Signals
- 2 Pairs of **Data (D)** and **Strobe (S)** in each direction

Data Strobe encoding

- At each bit time either the data or the strobe must change state
  - Never Both at the same time
- XORing data and strobe yields a synchronous clock







SpaceWire Characters

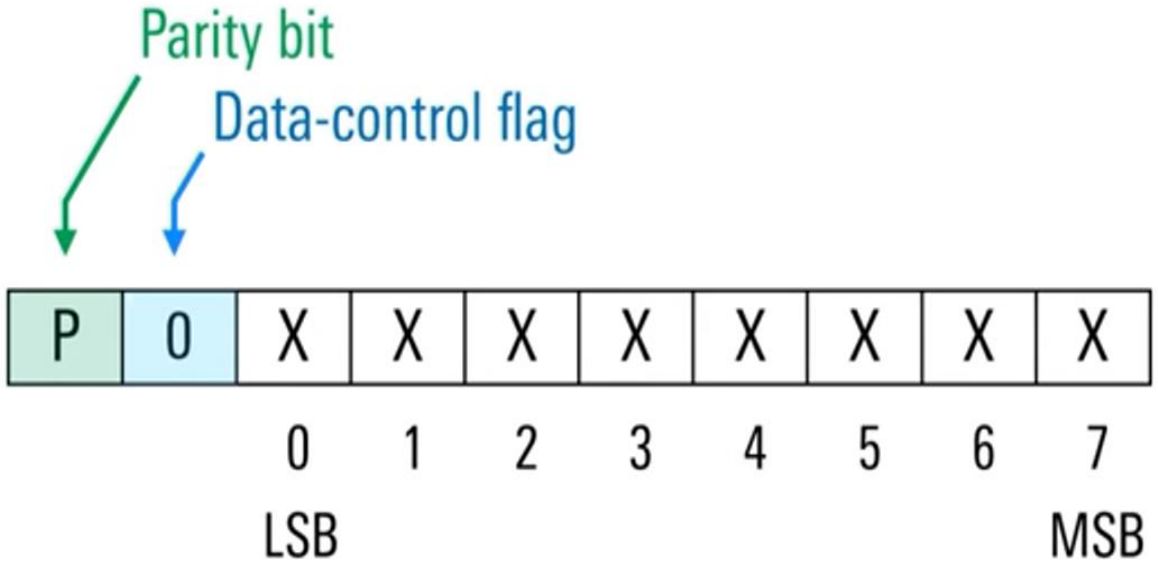
Data Character:

- 8 bits
- LSB is transmitted first

- Starts with **Parity Bit**
  - Used for error detection

- Followed by **Data-Control flag**
  - Set to 0 to indicate data character

- Data characters are used to:
  - Transfer Data info
  - Encode the destination address of a routed packet



SpaceWire Characters

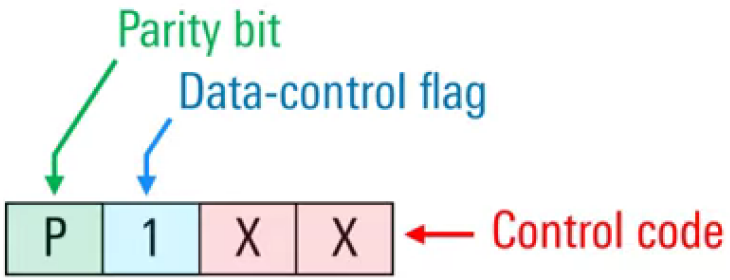
Control Characters:

- 4 bits
- LSB is transmitted first

- Starts with **Parity Bit**
  - Used for error detection

- Followed by **Data-Control flag**
  - Set to 1 to indicate control character

- Contains two-bit control code
  - **ESC** (escape) and **FCT** (flow control token) are used for link management
  - **EOP** (normal end of packet) and **EEP** (error end of packet) signal the end of a packet



P	1	1	1	ESC : Escape
P	1	0	0	FCT: Flow Control Token
P	1	0	1	EOP: (Normal) End of Packet
P	1	1	0	EEP: Error End of Packet

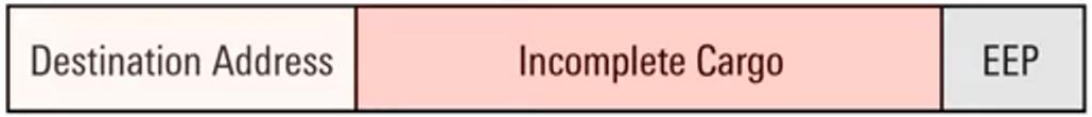
Packets

- **Packets** are composed of **Data Characters**

- **Destination Address**: Address of the path to the destination node
    - Not needed for p2p connection

- **Cargo:** One or more characters of user data
    - Can be any length

- **End of packet (EOP)**: Normal Packet end
- **Error end of packet (EEP):** Packet interrupted/ not completely transmitted
  (Cargo is incomplete and may be errored)



Control Codes

- **Escape (ESC):** control character used to create to longer codes.

- **Null Code:** used to
    - Initialize the link
    - Detect link disconnection
    - Sent to keep the link active

- **Time Code:** used for synchronization and time distribution

ESC : Escape

P	1	1	1

Null code

P	1	1	1	0	1	0	0

Time code

P	1	1	1	1	0	T	T	T	T	T	T	T	T
						0	1	2	3	4	5	6	7
						LSB							MSB

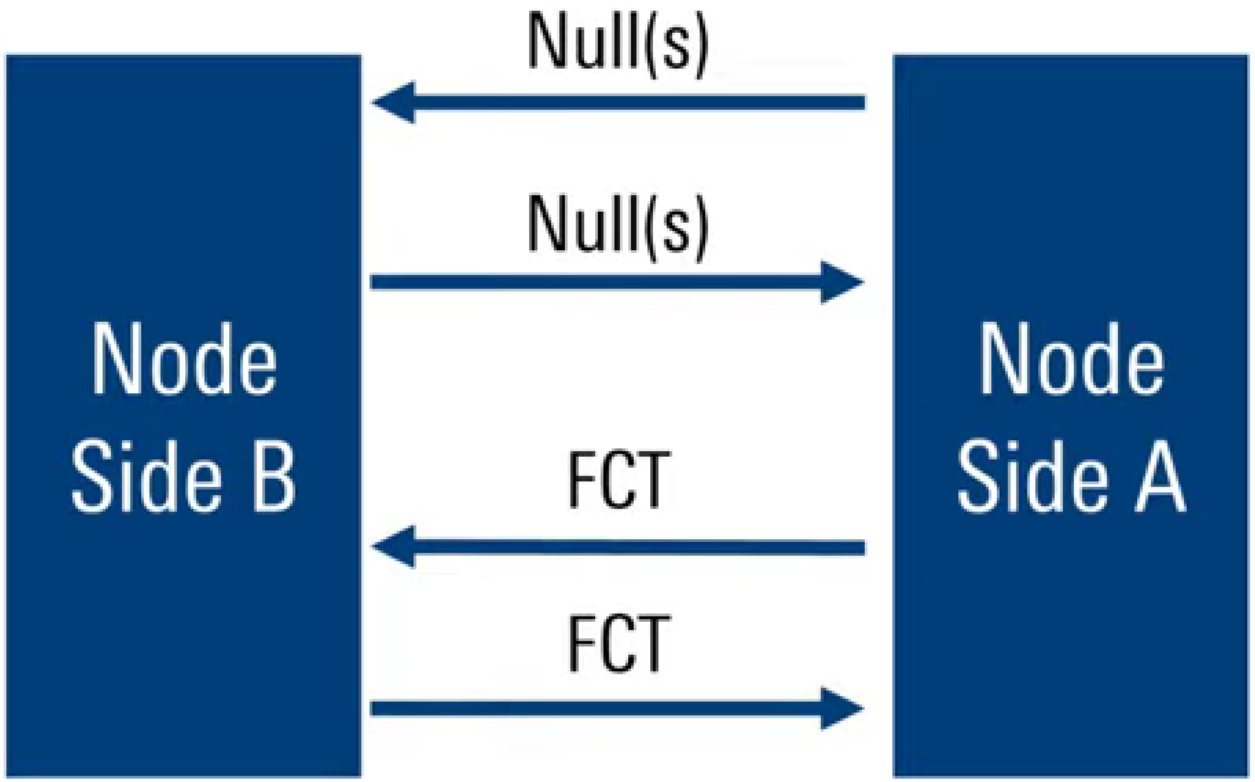
Link Initialization

- **Links** have to be **initialized** before data is sent

- **Link** must be **reinitialized** when an error is detected

- One end of the link **(Side A)** sends nulls
- The other end **(Side B)** syncs to these nulls and sends nulls back

- When **(Side A)** receives the nulls from **B,** it syncs to them
- **FCT(Flow Control Token)** can then be sent

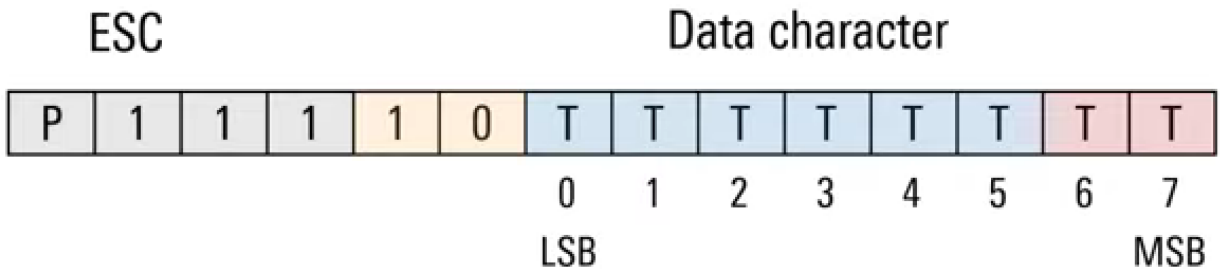


Time Distribution / Time Codes

- **SpaceWire** can distribute time info to all nodes on the network
    - Resolution in the nanosecond range

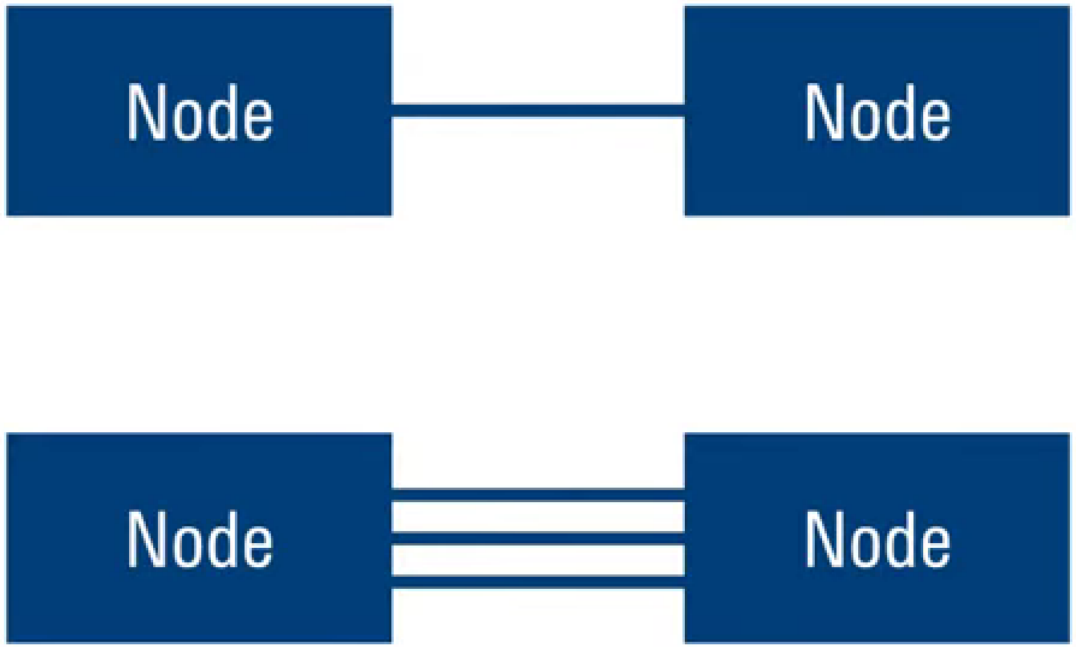
- This is done by broadcasting **Time-Codes**
    - Incrementing “ticks” synced to spacecraft time

- Combo of an **ESC** followed by a single **Data Character**
- **Time Code:**
    - 6 bit time count (0 to 5)
    - Two Control Flags (6 to 7) (Always zero)



Point-to-Point Links

- Nodes directly connected to each other
    - **Destination Address** field is not needed
- **Pros**
    - Simplicity
    - Low Power consumption
    - No contention/blocking
- **Cons**
    - Limited Flexibility
    - Lack of redundancy
- Multiple p2p connections can be used to share load

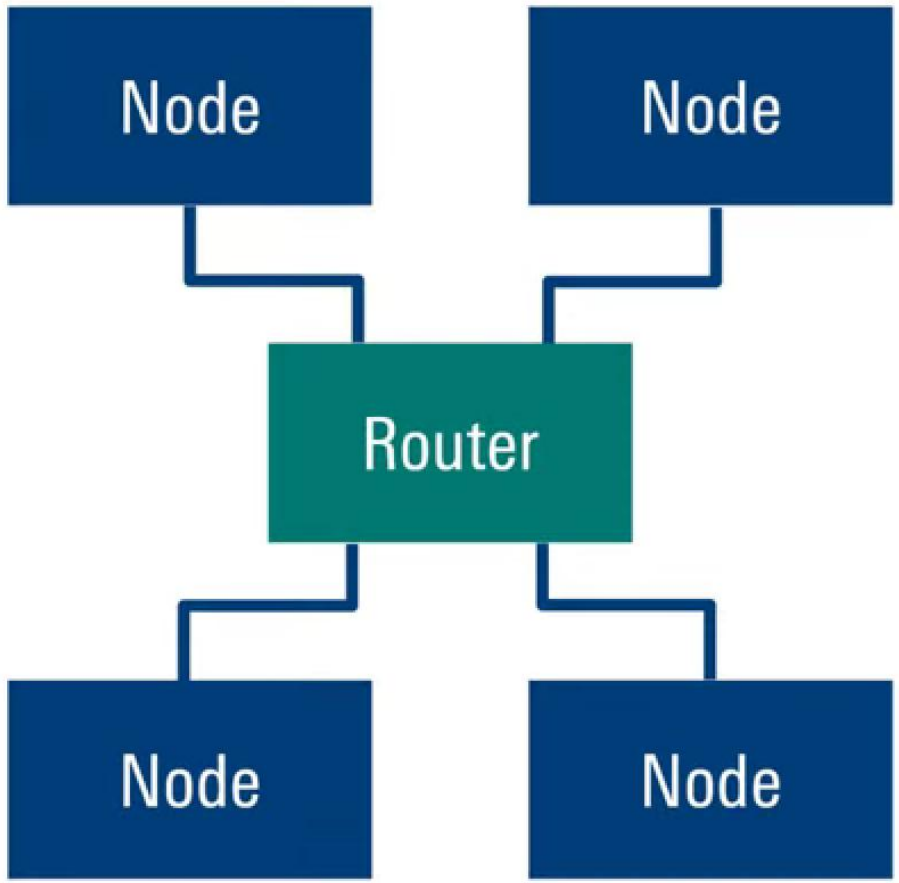


Routers

- Routing allows any nodes connected to the router to exchange information
- Routing provides architectural flexibility and reliability (when redundant paths exist.

- **Cons**
  - Power consumption
  - Single point of failure

- Routers use destination address to determine which link the packet should be routed to.
- **2 Types: Path Addressing/Logical Addressing.**

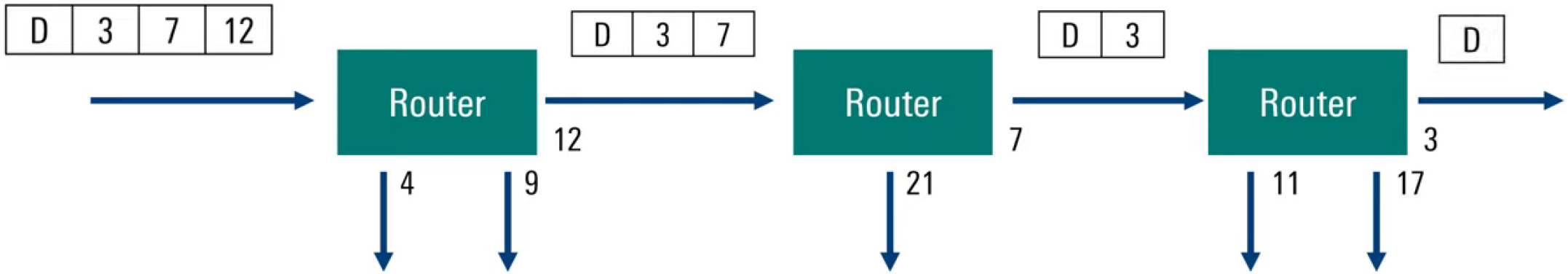


Path Addressing

- **Destination Address** is a series of data characters showing which exit port to take each successive router along the path to destination

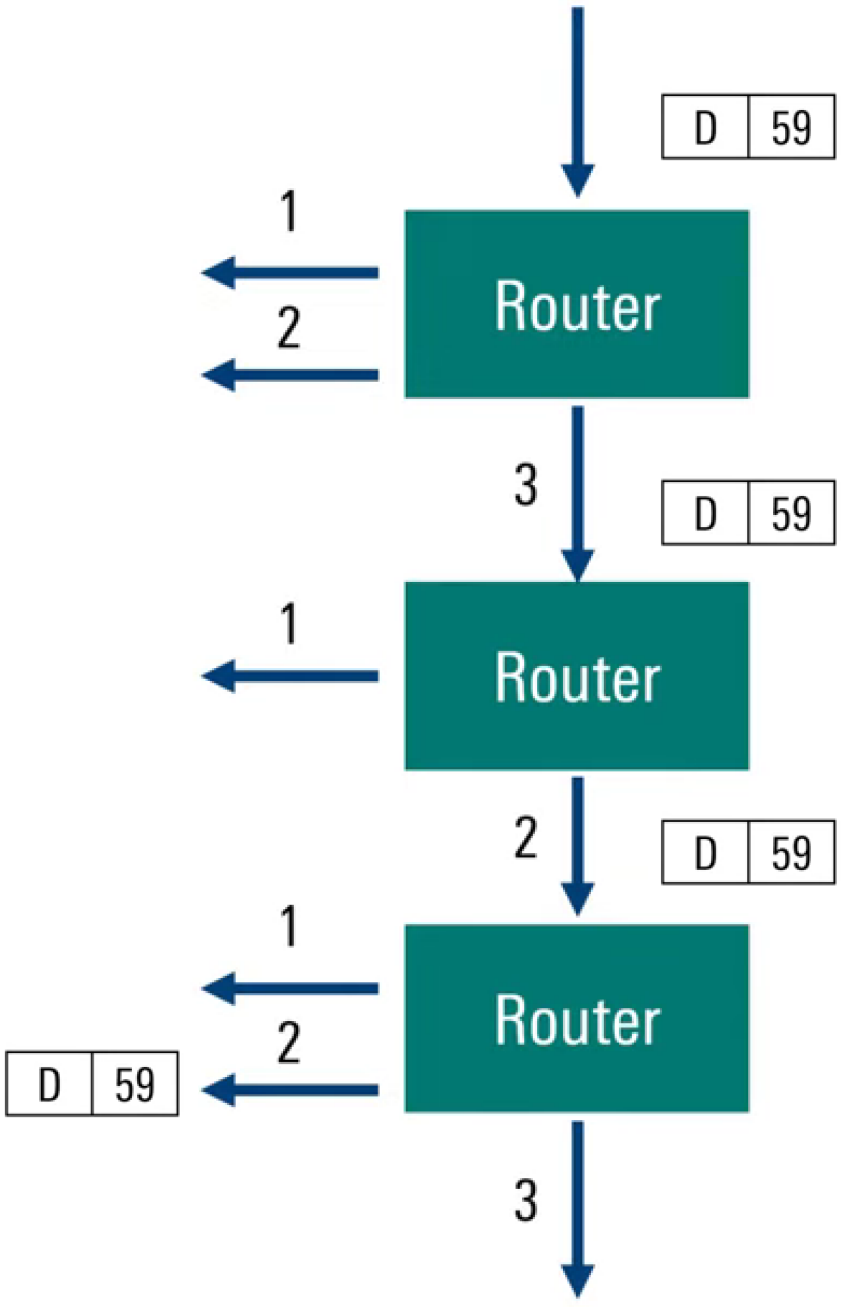
- Once the **leading character** has been **used** to route a packet, it is **stripped / discarded**

- Routers can have up to **31 external ports**
    - Each data character in the path address should be in the **range 0-31**
    - Routers don't need **Routing Tables**



Logical Addressing

- Only a **single character** is used for the **destination address**
    - **Not Deleted** as the packet passes through the router
- Each router uses its own **internal routing table**
- Destination Addresses must be in the range of 32-255
    - Avoids confusion with **Path Addressing** which use the 0-31 range



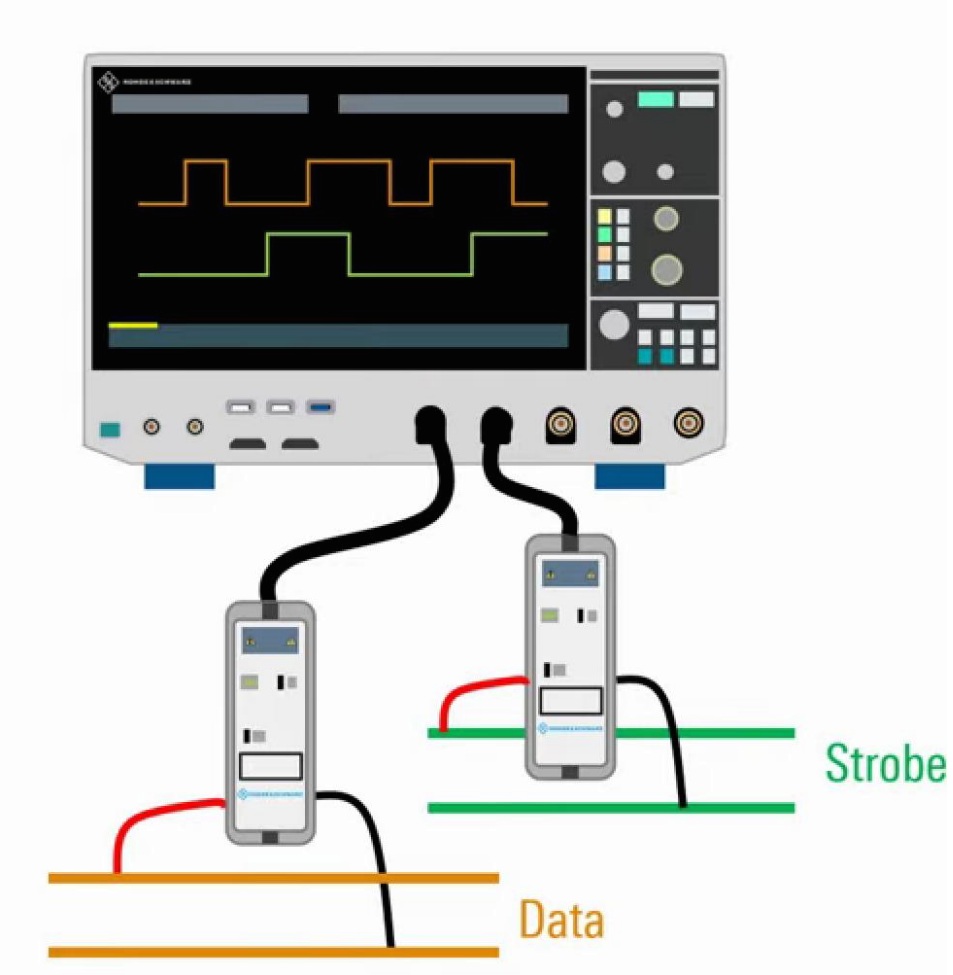
Address	Exit port
40-50	1
51-52	2
53-60	3

Address	Exit port
54	1
55-60	2

Address	Exit port
55-58	1
59	2
60	3

Decoding with Oscilloscope

- SpaceWire can be decoded using the appropriate decoder
- Two channels are needed
  + One channel for **Data,** one for **Strobe**
- Common applications include packet decoding, signal integrity/timing, and correlation between protocol and other events
- May also have protocol triggering and filtering
- Data can be exported in various formats



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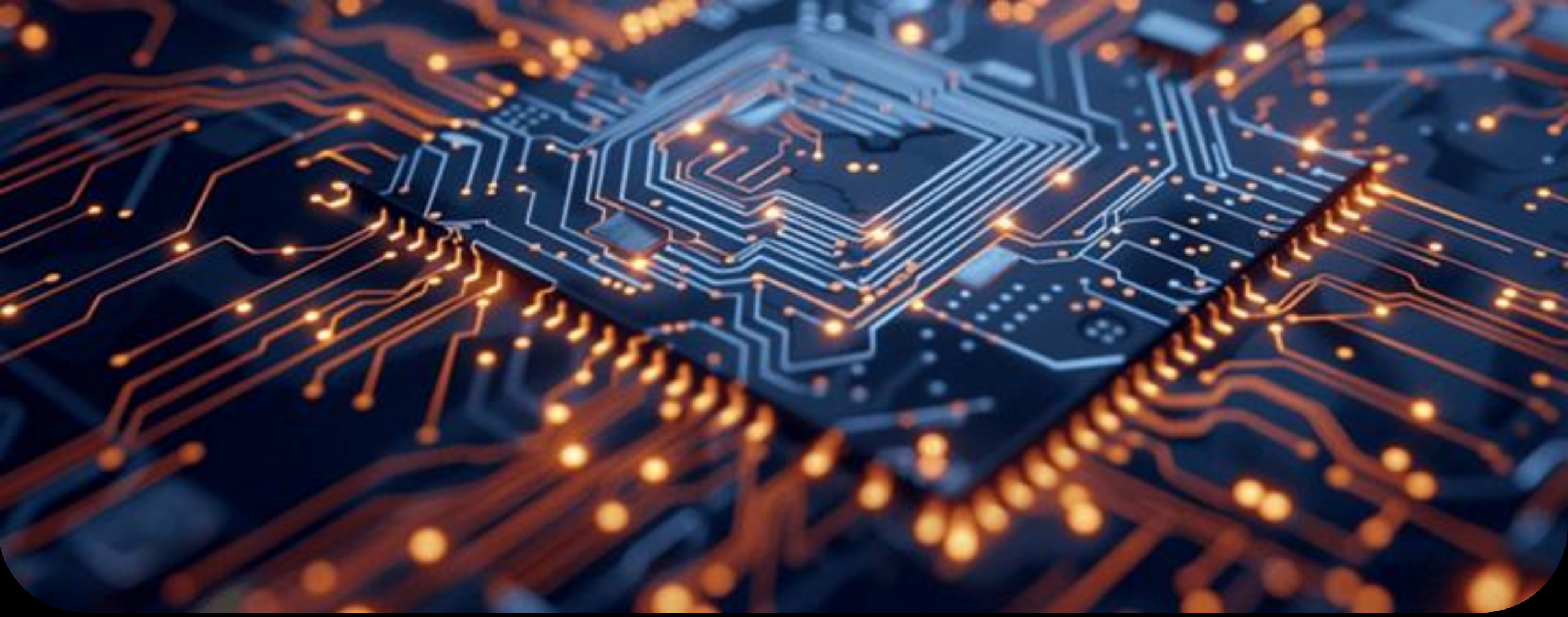
- Star-Dundee – SpaceWire's Guide https://www.star-dundee.com/wp-content/star_uploads/general/SpaceWire-Users-Guide.pdf



Thanks for ur time

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